Utilizing Machine Learning for Sport Data Analytics in Cricket: Score Prediction and Player Categorization

Suguna R
Department of Computer Science and Engineering
Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
Chennai, India
drsuguna@veltech.edu.in

Praveen Kumar Y
Department of Computer Science and Engineering
Vidhya Jyothi Institute of Technology
Hyderabad, India
praveen212.y@gmail.com

Suriya Prakash J
Department of Computer Science and Engineering,
JAIN (Deemed-to-be-University)
Bengaluru, India
suriyaprakash.j@jainuniversity.ac.in

Neethu P.S
School of Engineering and Technology
CHRIST (Deemed to be university)
Bengaluru, India
neethu.ps@christuniversity.ac.in

Kiran S
Department of Mathematics
Nitte Meenakshi Institute of Technology
Bangalore, India
kiran.s@nmit.ac.in

***Abstract*— Cricket is a popular sport with complex gameplay and numerous variables that contribute to team performance. In recent years, sports analytics has gained significant attention, aiming to extract valuable insights from large volumes of cricket data. Cricket has many fans in India. With a strong fan following, many try to use their cricket intuition to predict the outcome of a match. A set of rules and a points system govern the game. The venue and the performance of each player greatly affect the outcome of the match. The game is difficult to predict accurately as the various components are closely related. The CRR (Current Run Rate) approach is used to predict the final score of the first innings of a cricket match. Total points are calculated by multiplying the average number of runs scored in each over by the total number of overs. For ODI cricket, these methods are useless as the game can change very quickly regardless of the current run rate. The game may be decided by 1 or 2 overs. For more accurate score predictions, a system is needed that can more accurately predict the outcome of an inning. This research paper explores the application of machine learning techniques to predict scores and classify players based on their roles in the squad. The study utilizes a comprehensive dataset comprising various attributes of cricket matches, including player statistics, match conditions, and historical performance. Linear Regression, Logistic Regression, Naive Bayes, Support Vector Machines (SVM), Decision Tree, and Random Forest regression models are employed to predict scores. Additionally, player categorization is performed using a classification approach. The results demonstrate the effectiveness of machine learning techniques in enhancing performance analysis and decision-making in the game of cricket.**

Keywords*— *Cricket analytics, Cricket prediction, Machine Learning, Score prediction, Player Classification, Regression and Classification

I. INTRODUCTION

Cricket is a popular sport with a rich history and there are many factors that affect the outcome of a match. The advent of technology and the availability of large amounts of cricket data have opened up new opportunities for using machine learning techniques in sports analytics [15]. Harness the power of machine learning to gain deeper game insights, improve decision-making processes, and improve overall performance [5].

The main purpose of this research work is to study the application of machine learning techniques in cricket analysis, with a particular focus on outcome prediction and player classification. By using various machine learning algorithms such as Linear Regression, Logistic Regression, Naive Bayes, SVM, Decision Trees and Random Forest Regression Models, it is aimed to accurately predict match scores. Additionally, it is strived to categorize players into functional roles within the team so that it can better understand their contributions and abilities.

The scope of this research includes the analysis of comprehensive datasets covering various aspects of the cricket match, including player statistics, match conditions and past performance. The research focus is on the application of machine learning algorithms to predict outcomes given the large number of factors that influence the final outcome. Additionally, consider player classifications to identify different roles within a team to help with strategic planning and team selection. Using machine learning techniques, this research aims to contribute to the field of sports analytics in cricket and provide valuable insights to players, coaches, and cricket enthusiasts. The results have the potential to revolutionize match analysis, performance evaluation, and strategic decision-making in the context of this popular sport.

979-8-3503-4035-8/23/$31.00 ©2023 IEEE

II. RELATED WORK

Various prediction methods have been proposed in the past to calculate the final score and several criteria were discussed in classifying the players.

Ch Sai Abhishek [1] presented a model for predictive analysis of winners using machine learning techniques and implemented the above model to predict attractive outcomes. Each player's and team's history was observed, and the odds of winning against an opponent were determined based on this history. Various machine learning classifiers were trained and observed for this work, but Decision Trees and Random Forests proved to be more prominent, with an accuracy of 89.15%. Gagana S [5] implemented a machine learning-based approach for future analysis of T20 match results. The model predicts how many times a hitter will score with each ball by learning from observed data. This prediction was made using a variety of machine learning and data mining technologies. Using 90% training and 10% testing the accuracy score for Naive-Bayes, decision tree, random forest algorithms are 42.50% 82.52% and 90.88% respectively.

Daniel Mago Vistro [15] proposed a technique for predicting the winner based on machine learning and data analytics. The author suggested a model for predicting current matches, with this model focusing on numerous criteria such as a player's performance, team performance, venue, and weather conditions. Priyanka Sachi [10] studied years of IPL data, including player information, match site information, team information, and ball-to-ball information, to derive several conclusions that aid in the advancement of a player's performance. It focuses on calculating the outcomes of IPL matches using existing data mining approaches on both balanced and imbalanced datasets. It employs the Random Forest algorithm to estimate the score in each inning and, eventually, the winner of the match, taking into account criteria such as the number of wickets lost, the venue, and the toss.

B. Padmaja [6] emphasised the significance of machine learning in improving the performances of players and teams in various sports, as well as how wearable technology helps players understand their performance levels and future improvements. Monali Shetty [13] proposed a model that could select the top 11 players of the Indian cricket team. Each player's performance is determined by the type of field, the opposing team, the ground, and several other factors. The dataset for this model was developed using data from his trusted websites such as espn.com. This method is unique in that it provides a complete picture of a player's skill such as striking, bowling and fielding. A random forest algorithm is employed for predicting player performance. The model predicts hitters with 76 percent accuracy, bowlers with about 67 percent accuracy, and rounders with 95 percent accuracy. Vipul Punjabi [11] and his team used a simple Bayesian classifier to predict runs scored by batsmen and wickets scored by bowlers. Runs scored and wickets scored fall into different categories. The dataset used for this is taken from recordings of IPL matches. Very few input functions are required for this article, which severely limits the model possibilities.

Harshit Barot [3] mentioned a few essential factors such as team form and team strength in predicting the match outcome, apart from the traditional aspects such as the toss, the venue of the games, and so on. Furthermore, using Batting Index and Bowling Index, a novel analysis of batting and bowling has been provided. For match predictions, machine learning techniques such as SVM, Logistic Regression, Random Forest. The accuracy of the Decision Tree and Logistic Regression methods is over 87 percent and 95 percent, respectively. R. R. Kamble [7] proposed a system with two models. The first model estimates the team's score after he has 50 overs from the current state. The second approach predicts the odds of winning both sides before the game starts, via player selection.

Apurva Lawate [8] devised a model that can predict the first inning score as well as the winner of an IPL cricket match. The model's performance is decided by parameters such as the number of wickets taken in the previous 5 overs, the number of runs scored in the previous 5 overs, the number of overs, the overall score, and the number of wickets at the current ball. The proposed model incorporates data from IPL matches played between 2008 and 2019. This paper will show us how to forecast the expected score of the first inning while the game is still in progress. The Linear Regression approach is used to forecast the score. This model can account for 75.226 percent of the data. The model analyses data from the preceding five overs.

Pallavi Tekade [14] developed a machine learning model to predict the outcomes of its matches. Winning a cricket match is determined by several factors, including home ground advantage, previous performances on that venue, records at the same place, general experience of the players, record against a specific opposition, and the current form of the team and individual players. The primary factors that determine the outcome of a cricket match are discussed in this paper, as well as the regression model that best fits the data and delivers the most accurate forecasts. Nikhil Dhonge [4] proposed a model based on two strategies. The first is a prediction of team scoring, while the second is a prediction of team winning. In scoring prediction, linear regression, lasso regression, and ridge regression are employed, whereas SVC classifier, Random Forest Classifier is utilised owing to its high accuracy and consistency, ensuring that the intended anticipated result is accurate.

In order to forecast the outcome of a live T-20 match as well as the winner before play starts, an online tool for predictive analysis has been created [9]. The app also has a live score scraper that gathers precise match data and feeds it into the prediction algorithm. These predictions are made using algorithms like Random Forest Classifier and Multivariate Polynomial Regression, and Flask is utilised to construct the web application.

Researchers have looked into using machine learning techniques to forecast cricket results based on a range of historical and contextual variables [12]. A number of regression models that forecast a team's total runs scored in a limited-overs cricket match are constructed and assessed using the publicly accessible cricket dataset. Model selection is used in this research to compare and select among several regression techniques, feature engineering is used to extract and manipulate pertinent input variables, and performance assessment is used to gauge the models' robustness and accuracy.

One of the most significant cricket events is score prediction. The pursuing team gets ready in accordance with

the score projection once it is completed. The anticipated score for the first inning is determined using ML techniques and deep learning [2]. When developing a model, Ridge Regression and Linear Regression are employed, and several graphs are produced to identify the errors in order to assess the model.

III. METHODOLOGY

A. Feature Selection and Engineering

1) Features related to outcome prediction

When predicting cricket outcomes, the selection of relevant features plays an important role in understanding the factors that have a large impact on the final score. The feature selection process involves identifying attributes that show a strong correlation with the target variable (such as score). Common characteristics that are taken into account in predicting outcomes are:

Predicting the score of a cricket match requires complete ball-to-ball data for the match. This data may be obtained from the Cricsheet website https://cricsheet.org/ downloads/odis_male_csv2.zip. However, this data is limited to specific matches. Machine learning models need data on more matches to learn from data. So data was collected from individual matches and combined into a single data file using code written in Python. Cricsheet_ID is used to consolidate all available data into one data file. The final data set consisted of 316424 rows and 22 columns. The column details are shown in Table I.

TABLE I. LIST OF ATTRIBUTES IN THE DATASET

Feature	Description
Season	describe the season in which the games are played.
Start_date	shows information about the date the game is played.
Venue	provides information about the name of the location where the game is played.
Innings	provides information on whether it was the first or his second. It contains two values 1 and 2.
Ball	provides information about game progress.
Batting team	name of batting teams
Bowling team	name of the bowling team
Striker	name of the batsman facing the current ball
Non striker	name of the batsman at the crease
Bowler	name of the current bowler
Runs_off_bat	the number of runs the batsman has scored [0-6]
Extras	Extras occurred on that particular ball
Wides	wide ball that has been bowled on that particular ball [0-5]
No balls	no ball that has been bowled on that particular ball [0-5]
Byes	runs taken from the miss fielding
Leg byes	runs taken when the ball gets deflected after striking the batsman's pads.
Penalty	Penalty will be given if the ball bowled is a dead ball
Wicket_type	wicket fallen on that particular delivery
Player dismissed	player who is dismissed on the current ball
Other wickets	wickets that fell due to mistakes of the batsman such as retired hurt
Other player dismissed	players who are dismissed because of personal issues
Total_score	total score the team has scored till that particular ball
Final Score	total score the team has scored in its entire innings
Striker_score	runs scored by the batsman who is in strike
Non_striker_score	runs scored by the batsman who is in non-striker's position
Wickets	number of wickets that have fallen
Last_5_overs_score	used to predict the final score

2) Features related to player classification

The features and statistics of players that is the most vital in player classification have been calculated from the dataset. The columns used for player classification are shown in Table II.

TABLE II. FEATURES USED FOR PLAYER CLASSIFICATION

Feature	Description
Player	name of the player
Innings	number of innings the players has played
Runs	number of runs the player has scored throughout his career
Average	average of that particular player runs and innings
Strike rate	Strike rate of the particular player
Wickets	number of wickets that player has taken throughout his career
Economy	Overs bowled in a game
Bowling average	Computed using runs concealed and wickets taken by bowler
Bowling strike rate	describe how many balls bowler throws for each wicket he takes
Batting basra	aggregate of batting average and strike rate of a player
Bowling basra	Ratio of wickets taken by the bowler to number of matches he has

3) Machine Learning Techniques

a) Linear Regression

Linear Regression is a popular machine learning algorithm used for predicting continuous variables, making it suitable for score prediction in cricket. It establishes a linear relationship between the input features and the target variable (score) to make predictions. The model learns the coefficients of the linear equation through a training process and applies them to new data for score prediction.

In matrix notation, the linear regression equation for single and multiple can be represented as in (1):

$$Y=X.\beta+\in \quad (1)$$

where

- Y is a vector of observed values of the dependent variable
- X is a matrix of independent variables
- β is a vector of coefficients (including the intercept)
- ϵ is a vector of error terms

b) Logistic Regression

While Linear Regression is suitable for predicting continuous variables, Logistic Regression is specifically designed for binary classification tasks. In the context of cricket analytics, Logistic Regression can be utilized to predict binary outcomes, such as whether a team will win or lose a match based on various features. It estimates the probability of the binary outcome using a logistic function. Logistic regression equation can be represented as in (2).

$$P(y=1|X)=1/(1+e^{(-(X.\beta))}) \quad (2)$$

where

- P(y=1|X) is a vector of predicted probabilities for class 1
- X is a matrix of independent variables
- β is a vector of coefficients (including the intercept)

- ϵ is a vector of error terms

c) Naive Bayes

Naive Bayes is a probabilistic classification algorithm commonly used for text classification tasks. However, it can also be employed in cricket analytics for player categorization. By considering the statistical independence assumption between features, Naive Bayes calculates the probability of a player belonging to a particular category based on their attribute values. It assigns players to the category with the highest probability. Multinomial Naive Bayes estimates the conditional probability P(y|x) using Bayes' theorem and the assumption of conditional independence as shown in (3):

$$P(x) = \frac{P(y).P(y)}{P(x)} \quad (3)$$

where

- P(y|x) is the posterior probability of class y given features x
- P(x|y) is the likelihood of observing the features x given class y
- P(y) is the prior probability of class y
- P(x) is the evidence, the probability of observing the features x

d) Support Vector Machines (SVM)

Support Vector Machines are powerful algorithms used for both classification and regression tasks. In cricket analytics, SVM can be employed for score prediction and player categorization. SVM aims to find the optimal hyperplane that best separates data points of different classes. It can handle both linear and non-linear relationships between features and the target variable through the use of kernel functions. The objective is to minimize the sum of squared errors between the predicted values and the actual target values as represented in (4).

$$\frac{1}{2}||w||^2 + C \sum_{i=1}^{n} (\varepsilon_i + \varepsilon_i^*) \quad (4)$$

where

- w is the weight vector
- ε_i,ε_i^* are slack variables representing the deviation of the predicted values from the actual values.
- C is a regularization parameter

e) Random Forest Regression

Random Forest Regression is an ensemble learning technique that combines multiple decision trees to improve prediction accuracy. In the context of score prediction in cricket, Random Forest Regression can be utilized by aggregating the predictions of multiple decision trees to provide a more robust and accurate score prediction. It helps in reducing overfitting and capturing complex relationships between features and the target variable. . The final Random Forest regression prediction is the average (or weighted average) of the predictions from all individual decision trees as represented in (5).

$$\hat{y} = \frac{1}{M} \sum_{i=1}^{M} f_i(X) \quad (5)$$

where

- ŷ is the predicted target value for a given set of features X
- f_i (X) is the the prediction of the i-th decision tree in the Random Forest

f) XGBoost (eXtreme Gradient Boosting) Regressor

The XGBoost (eXtreme Gradient Boosting) Regressor is an implementation of gradient boosting that has gained popularity for its efficiency and effectiveness. It enhances the traditional gradient boosting algorithm by incorporating several key optimizations and advanced features. It has been successfully applied to cricket score prediction and player performance analysis.

The XGBoost Regressor aims to create an additive model of decision trees, where each new tree is added to correct the errors of the previous trees and can be represented as (6).

$$\hat{y} = \sum_{m=1}^{M} \gamma_m f_m(X) \quad (6)$$

where

- ŷ is the predicted target value for the features X
- f_m (X) is the prediction of the m-th decision tree
- γ_m is the weight (scaling factor) associated with the m-th tree's prediction

IV. Results and Discussion

A. Score Prediction

1) Model Training and Evaluation

To predict cricket scores using machine learning techniques, the collected and preprocessed dataset is split into training and testing subsets. The training set is used to train the machine learning models, while the testing set is used to evaluate their performance. The models are trained by feeding them with input features (e.g., historical performance, match conditions) and the corresponding target variable (scores). Fig 1 shows the process flow in score prediction.

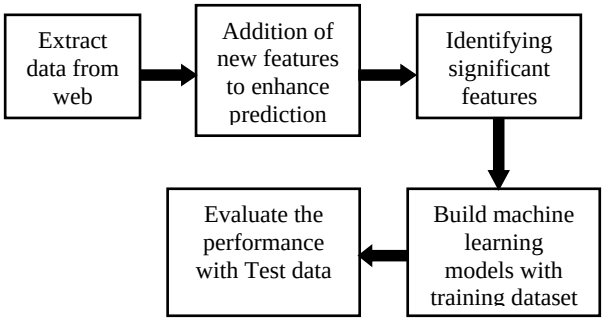


Fig. 1 Process flow in score prediction

2) Performance Metrics

Various performance measures are utilized to assess the accuracy and efficacy of the score prediction algorithms. These metrics are as follows:

a) Mean Absolute Error (MAE): This statistic calculates the average absolute difference between anticipated and actual results. Lower MAE values indicate more precision.

b) Mean Squared Error (MSE): This metric computes the average squared difference between anticipated and actual scores. MSE is extensively utilised in

regression problems because it provides more weight to larger errors.

c) Root Mean Squared Error (RMSE): The square root of MSE, RMSE measures the average prediction error in the original score unit.

d) R-squared (R2): R-squared calculates the proportion of the target variable's variance that can be explained by the model. A higher R-squared value indicates a better fit of the model to the data.

B. Comparative Analysis of Models

After training and evaluating multiple machine learning models, a comparative analysis is conducted to determine the most effective model for score prediction in cricket. The models' performance metrics are compared, and their strengths and weaknesses are evaluated based on performance metrics.

Table III summarizes the predicted and actual score of the different regression models.

TABLE III. PERFORMANCE OF REGRESSION MODELS

Models	Predicted	Actual	Models	Predicted	Actual
Random Forest	351.3	352	SVM	314.81	352
	284.77	251		244.95	251
	116.8	116		168.59	116
	260.65	257		246.89	257
	310.29	314		234.84	314
Linear Regression	334.8	352	Gradient Boost Regressor	324.81	352
	221.87	251		244.95	251
	178.8	116		168.59	116
	229.11	257		246.89	257
	246.38	314		246.84	314

The performance assessments of models on different metrics are shown in Fig 2.

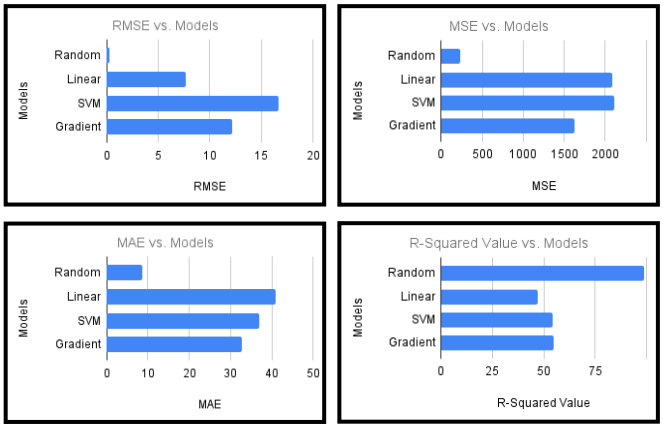


Fig. 2 Performance of models on different metrics

The analysis reveals that Random forest regression models perform well compared to other models.

After training the models, it must validate them by feeding the inputs of specific matches into the system such as the location of the match, the details of the innings, first/second innings, the teams that are playing the match, the batsmen who are at the crease, the striker and non-striker and their runs, the bowler who is currently bowling, the number of wickets, runs scored in the last five overs, and the current score. The demonstrated output is shown in Figure 3.

```
Enter yes to predict score, else noyes
Enter venue Old Trafford
Enter Innings 1
Enter ball 11.4
Enter batting team India
Enter bowling team West Indies
Enter Striker MS Dhoni
Enter non Striker V Kohli
enter striker runs 32
Enter non-striker runs 28
Enter bowler O Thomas
enter no of wickets fallen 2
Enter last 5 overs score 25
Enter current total score 93
Linear Regeression Prediction score: [287.43088776]
SVM Prediction score: [313.83285094]
Gradient Boosting Prediction score: [281.66402655]
Random Forest Prediction score: [228.61]
```

Fig. 3 Input for Score prediction

C. Classification of Player

For the classification of players, a new dataset has been created from the existing dataset. The dataset usually contains statistics of players. To the existing dataset, two new features have been added namely batting basra and bowling basra which would help us classify the allrounders as batting allrounders and bowling allrounders.

Classification algorithms such as logistic regression, naïve Bayes, SVM classifier, random-forest-classifier and XG Boost classifier are used for the study. The results for the algorithms are shown in Fig 4.

```
Logistic Regression Accuracy:  0.9135135135135135
SVM Accuracy:  0.9513513513513514
Naive bayes Accuracy:  0.8810810810810811
Random Forest Accuracy:  0.9891891891891892
XGBoost Classifier Accuracy:  0.9945945945945946
```

Fig. 4 Performance of Classification Algorithms

The user must provide the player's statistics as input for classification. The statistics include the number of innings played, the number of runs scored, the strike rate of the player, the number of wickets taken, the bowling economy, bowling average, bowling strike rate, batting basra, and bowling basra. The algorithms then divide the players into four categories: batsman, bowler, batting allrounder, and bowling allrounder. Figure 5 shows the input and player classification result of different classifiers.

```
Enter yes to classify a player ,else noyes
Enter number of innings played by the player 21
Enter runs of the player 334
Enter player strike rate43
Enter number of wickets taken by the player18
Enter bowling economy3.2
Enter bowling average37
Enter bowling striker rate42
Enter bowlingbasra of the player 52
Enter battingbasra of the player64
Linear Regeression Classifier: bowler
SVM Prediction classifier: bowling_allrounder
Naive Bayes classifier: bowler
Random forest  classifier: bowler
```

Fig. 5 Player Classification

V. CONCLUSION

The field of sports analytics is one that is fast developing and has enormous potential to improve player and organization performance. This technology allows trainers to find areas where athletes may grow and help them enhance their talents by systematically identifying and resolving deficiencies. Additionally, projecting their anticipated scores, helps teams to make tactical changes.

However, a noteworthy obstacle in this field is the shortage of appropriate online resources. In order to get over this limitation, the research set out to gather match data from multiple sources and combine it into a comprehensive dataset in order to forecast overall scores. Interestingly, the primary dataset does not have the necessary statistics needed to classify players. A new dataset has been created that includes important statistics from the earlier mentioned dataset in order to overcome this limitation. Regression and classification methods were both included in the proposed strategy. Players were classified using classification techniques, and scores were predicted using regression models. The introduction of new terms, including "batting basra" and "bowling basra," made player classification easier.

After experimenting with four regression algorithms in the investigation, it was discovered that the Random Forest Regressor produced the most accurate predictions. Five different approaches were investigated for player classification, with XGBooster turning out to be the most successful. The potential of sports analytics to maximize the performance of players and teams is highlighted by these studies.

REFERENCES

[1] C. S. Abhishek, K. V. Patil, P. Yuktha, M. K. S, and M. V. Sudhamani, "Predictive Analysis of IPL Match Winner using Machine Learning Techniques," Int. J. Innov. Technol. Explor. Eng., vol. 9, no. 28, pp. 430-435, 2019. DOI: 10.35940/ijitee.b1043.1292s19.

[2] R. Ahmed et al, "First inning score prediction of an IPL match using machine learning," AIP Conf. Proc., vol. 2555, 020018, 2022. DOI: 10.1063/5.0108928.

[3] H. Barot, A. Kothari, P. Bide, B. Ahir, and R. Kankaria, "Analysis and Prediction for the Indian Premier League 2020," in International Conference for Emerging Technology (INCET), Belgaum, India, 2020, pp. 1-7. DOI: 10.1109/INCET49848.2020.9153972.

[4] N. Dhonge et al., "IPL Cricket Score And Winning Prediction Using Machine Learning Techniques," Int. Res. J. of Modernization in Engineering Technology and Science, vol. 3, no. 5, 2021.

[5] S. Gagana, "A Perspective on Analyzing IPL Match Results using Machine Learning," Int. J. for Sci. Research & Development, vol. 7, no. 3, pp. 1476-1479, 2019.

[6] P. S. Harsha Vardhan Goud, Y. Mohana Roopa, and B. Padmaja, "Player Performance Analysis in Sports: with Fusion of Machine Learning and Wearable Technology," in 3rd International Conference on Computing Methodologies and Communication (ICCMC), Erode, India, 2019, pp. 600-603. DOI: 10.1109/ICCMC.2019.8819815.

[7] Kamble et al., "Cricket Score Prediction Using Machine Learning," Turkish J. of Computer and Mathematics Education, vol. 12, no. 1S, pp. 23-28, 2021.

[8] A. Lawate et al., "Cricket Analysis and Prediction of projected Score and Winner using Machine Learning," Int. J. of Adv. Research in Computer and Comm. Engg., vol. 10, no. 2, 2021.

[9] E. Mundhe, I. Jain, and S. Shah, "Live Cricket Score Prediction Web Application using Machine Learning," in 2021 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON), Pune, India, 2021, pp. 1-6. DOI: 10.1109/SMARTGENCON51891.2021.9645855.

[10] S. Priyanka, "Prediction of Indian Premier League-IPL 2020 using Data Mining Algorithms," Int. J. for Research in Applied Science and Engineering Technology, vol. 8, pp. 790-795, 2020.

[11] V. Pujbai, R. Chaudhari, D. Pal, K. Nhavi, N. Shimpi, and H. Joshi, "A survey on team selection in game of cricket using machine learning," Int. Res. J. of Engineering and Technology, vol. 6, no. 11, 2019.

[12] P. Sanjeeva et al, "Automated Cricket Score Prediction," E3S Web of Conferences, vol. ICMPC 2023, 01053, 2023. DOI: 10.1051/e3sconf/202343001053.

[13] M. Shetty, S. Rane, C. Pandita, and S. Salvi, "Machine learning-based Selection of Optimal sports Team based on the Players Performance," in 2020 5th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2020, pp. 1267-1272. DOI: 10.1109/ICCES48766.2020.9137891.

[14] P. Tekade, K. Markad, A. Amage, and B. Natekar, "Cricket Match Outcome Prediction Using Machine Learning," Int. J. of Adv. Scientific Research and Engineering Trends, vol. 5, no. 7, 2020.

[15] D. M. Vistro, F. Rasheed, and L. G. David, "The Cricket Winner Prediction With Application Of Machine Learning And Data Analytics," Int. J. of Scientific & Tech. Research, vol. 8, no. 9, pp. 985-990, 2019.